TASK 2: EXCEPTION HANDLING

Python Programming Practical Report — 20 Marks

1. Aim

To write a Python program that handles common runtime errors, including invalid user input, division by zero, and a file-not-found error, using appropriate try and except blocks.

2. Problem Statement

Develop a program that demonstrates exception handling for three common situations: entering a non-numeric value when an integer is expected, dividing a number by zero, and attempting to open a file that does not exist.

3. Concepts Used

Exception	Purpose
ValueError	Handles invalid input when converting text to a number.
ZeroDivisionError	Handles attempts to divide a number by zero.
FileNotFoundError	Handles attempts to open a file that is missing.
try / except	Runs potentially risky code and handles errors gracefully.

4. Source Code

```
# Task 2: Exception Handling

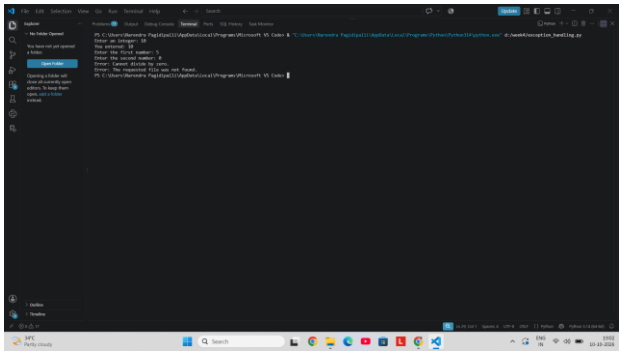
# 1. Handle invalid user input
try:
    num = int(input("Enter an integer: "))
    print("You entered:", num)
except ValueError:
    print("Error: Please enter a valid integer.")

# 2. Handle division by zero
try:
    a = int(input("Enter the first number: "))
    b = int(input("Enter the second number: "))
    result = a / b
    print("Division result:", result)
except ZeroDivisionError:
    print("Error: Cannot divide by zero.")
except ValueError:
    print("Error: Please enter valid numbers.")

# 3. Handle file not found
try:
    file = open("unknown.txt", "r")
    print(file.read())
    file.close()
```

```
except FileNotFoundError:
    print("Error: The requested file was not found.")
```

5.Output



6. Project Explanation

1. The try block contains code that may cause a runtime error.
2. ValueError is handled when the user enters text instead of an integer.
3. ZeroDivisionError is handled when the second number is zero.
4. FileNotFoundError is handled when unknown.txt does not exist in the program's current folder.
5. Meaningful messages are printed so the program can explain the problem instead of showing an unhandled traceback.

7. Challenges Faced

A common challenge is predicting which errors may occur and handling each with the correct exception type. This program uses separate try and except blocks for each situation, making the code easier to understand and test.

8. Conclusion / Result

The program demonstrates exception handling for invalid user input, division by zero, and file not found. Appropriate try and except blocks display meaningful error messages for each case.